Effects of Lithium therapy on blood glucose and insulin levels in individuals with bipolar disorder: a systematic review

Kanter-Eivin, David[1]; Blinkov, Konstantin[2]; MacDonald, Sadie[2]; Gibson, Jada[2]; Pimenta, Sarah[2]; Teopiz, Kayla[3]; Kwan, Angela T.H.[3,5]; Le, Gia Han[3,6]; Wong, Sabrina[3,7]; Rosenblat, Joshua D.[5,6,7]; McIntyre, Roger S.[3,4,7]

[1] Michael G. DeGroote School of Medicine, McMaster University

[2] Faculty of Health Sciences, McMaster University

[3] Brain and Cognition Discovery Foundation

[4] Department of Psychiatry, University of Toronto

5 Faculty of Medicine, University of Ottawa, Ottawa, ON, Canada

6 Institute of Medical Science, University of Toronto, Toronto, Canada

7. Department of Pharmacology & Toxicology, University of Toronto, Toronto, Canada

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Abstract

Background: Lithium is the gold-standard pharmacotherapy for both acute episodes and long-term management of bipolar disorder (BD). Lithium treatment is associated with metabolic changes including weight gain; however, the exact mechanism for this is poorly understood. This systematic review assesses the effects of lithium on glucose, glucose-tolerance, and insulin in persons with BD.

Methods: A systematic search for studies was conducted on PubMed, MEDLINE, Embase, CENTRAL, APA PsychINFO, Scopus and Web of Science. Primary studies assessing validated metrics of serum glucose or insulin in bipolar patients receiving lithium treatment were included.

Results: 16 studies were included in this review. Fasting plasma glucose (FPG) was the most thoroughly studied metabolic biomarker, and lithium use was not associated with changes to FPG. There was some evidence that lithium treatment increases glucose tolerance in the short-term, but this effect decreases over time and is transient. Studies examining other glucose metrics and insulin had less concordance with one another.

Limitations: There is a lack of a consistent criteria for BD across the reviewed studies. Many included studies lack controls and do not stratify for important metabolic factors such as body mass index (BMI) and age.

Conclusions: Lithium does not directly impact FPG in persons with BD. Evidence supports the metabolic safety of lithium with respect to FPG, and any observed changes are likely secondary to changes in BMI, age, or mood status. Further research is needed to understand lithium's impact on other metabolic parameters.

Introduction

Bipolar disorder (BD) is a mood disorder characterized by episodes of manic mood dysregulation followed by depressive episodes. BD is classified into bipolar type I (BD1), characterized by severe manic episodes followed by a depressive period, and bipolar type II (BD2), characterized by periodic hypomanic and depressive episodes (McIntyre et al., 2020). BD has a lifetime prevalence of 1-2% globally (Merikangas et al., 2011). Compared to healthy individuals, individuals with BD have a 10-20 year shorter life expectancy on average, and significantly higher rates of comorbid chronic and non-communicable diseases such as cardiovascular disease (McIntyre et al., 2020).

Acute treatment for BD typically focuses on the stabilization of manic and/or depressive episodes, while maintenance treatments aim to reduce the frequency and severity of symptoms, prevent recurrence and optimize health-related quality of life (Geddes and Miklowitz, 2013). Pharmacological treatment of mania typically involves one or more of lithium, antipsychotics (e.g. quetiapine, aripiprazole) and anticonvulsants (e.g. valproate, carbamazepine) (Geddes and Miklowitz, 2013).

Despite being introduced over 50 years ago, lithium remains the best-established long-term treatment for bipolar disorder (Geddes and Miklowitz, 2013). Lithium's efficacy has been repeatedly demonstrated by randomized-controlled trials (RCTs), showing lithium maintenance therapy to reduce manic and depressive relapse risk by 38% and 28%, respectively (Geddes et al., 2004; Severus et al., 2014). In addition, lithium is also established as an anti-suicidal treatment, with evidence supporting an approximate 60% reduction in risk of suicide (Cipriani et al., 2005; Geddes and Miklowitz, 2013; McIntyre et al., 2020). In contrast to other agents which are primarily effective against mania, lithium is effective against both manic and depressive episodes (Cipriani et al., 2005; McIntyre et al., 2020). Lithium is often administered in combination with an anticonvulsant (AC) or antipsychotic (AP), which has demonstrated greater efficacy than either lithium or antipsychotics alone (Cipriani et al., 2005).

While lithium remains a first-line treatment for BD (Geddes and Miklowitz, 2013; McIntyre et al., 2020), its benefits are restricted by a low therapeutic index and significant adverse effects including metabolic changes (i.e. weight gain), hypothyroidism, cognitive impairment, and teratogenicity (McIntyre et al., 2020). The metabolic effects of lithium is a major area of ongoing and previous research, which largely has yielded conflicting findings on lithium's contribution to metabolic dysfunction. Research is complicated by the difficulty of differentiating the effects of lithium from the underlying effects of BD, which is associated with a higher prevalence of type II diabetes mellitus (T2DM), obesity, metabolic syndrome, and metabolic dysfunction associated fatty-liver disease (Jawad et al., 2023; Liu et al., 2022). Additionally, lithium is frequently administered as a combination therapy with atypical antipsychotics (AAP) or AC, which are also associated with changes in weight and metabolic parameters (McIntyre et al., 2024). Given the numerous factors which underlie the pathogenesis of metabolic disease, many of which are poorly understood, these conflating variables have made it difficult to identify the *specific* effects of lithium on metabolic parameters, such as blood glucose levels.

While lithium's relationship to metabolic syndrome has been the focus of numerous primary studies and reviews, far fewer studies have specifically assessed the effects of lithium on glucose and glucose-sensitivity markers, such as insulin and glucose tolerance tests (GTT). This systematic review aims to elucidate the independent effects of lithium on plasma glucose, blood insulin levels, and insulin resistance, through evaluating extant literature reporting on lithium-glucose or lithium-insulin relationships.

Methods

Search Strategy

This review focuses on examining the effects of lithium therapy on glucose and insulin homeostasis in persons with BD. A systematic search for articles was conducted across PubMed, MEDLINE, Embase, CENTRAL, APA PsychINFO, Scopus, and Web of Science. The keywords used for the search were:

(bipolar OR bipolar disorder OR bipolar depression OR mania OR hypomania) AND (lithium OR lithium monotherapy) AND (glucose OR insulin OR fasting plasma glucose OR fasting insulin OR serum glucose OR serum insulin OR insulin signaling OR insulin sensitivity).

After removing duplicates (automatically and manually), title and abstracts were screened by two independent reviewers (any two of DK, JG, KB, SP) against the selection criteria. In cases where conflict of study inclusion occurred between the two reviewers, a third reviewer (DK or JG) determined the inclusion status. Studies which were included based on their title and abstract then underwent full-text review by two independent reviewers.

Inclusion/Exclusion Criteria

We included studies that i) were primary, cross-sectional, prospective, or retrospective; ii) followed patients with BD or documented manic-depressive symptoms; iii) followed patients treated with lithium monotherapy or combinations of lithium and other medications (e.g. valproate, carbamazepine); iv) measured fasting plasma glucose (FPG), glycated RBC NMR, random glucose, serial glucose measurement (mean or nadir), oral glucose tolerance test (GTT), or insulin.

We excluded studies which i) exclusively relied on pre-clinical data or animal trials; ii) did not quantitatively report measures of plasma glucose or insulin levels (e.g. brain imaging studies); iii) focused on out-of-scope metabolic parameters relating more broadly to metabolic syndrome and diabetes, such as serum lipid levels, HbA1c, and obesity; iv) were not peer reviewed; or v) were not primary research articles. We also excluded case reports, which often provide a complex clinical picture that is not readily generalizable, and where it is difficult to isolate the effects of lithium. We did not exclude studies based on their date of publication.

Information Extraction

Relevant findings were extracted using a piloted form. Key data included: i) first author and publication year; 2) study design; 3) study location; 4) metrics assessed (i.e. FPG, random glucose, insulin); 5) participant characteristics (age, gender, BMI) and sample sizes; 6) criteria used for making a bipolar diagnosis (if applicable); 7) intervention (duration, dosage, adjunctive therapies); 8) fasting, random, mean, and nadir glucose, as well as insulin levels, if reported. Each article was extracted by one author (one of DK, JG, KB, SP), and the extracted data was verified by a separate author (one of DK, KB, SM).

Results

In total, 2870 studies were returned from the search, with 1360 remaining after the removal of duplicates. Based on title and abstract screening, 53 articles were selected for full-text review, and 16 studies were included after full-text screening (Figure 1). 10 studies were prospective, 3 were retrospective, and 3 were cross-sectional in design. 7 of the studies were conducted in the United States, 6 in Europe, 1 in Canada, 1 in Turkey, and 1 across several countries. Between the studies, participants ranged between 15 and 75 years old. Multiple metrics were used to measure glucose and insulin in study patients. An overview of the included articles is provided in Table 1.

Studies of Fasting Plasma Glucose

Five studies investigated lithium's effect on the fasting plasma glucose (FPG) of persons with BD, with the majority finding no significant effect. Two studies, Kuperberg et al. (2022) and Köhler-Forsberg et al. (2023), conducted pooled analyses of two 24-week randomized trials (Bipolar CHOICE and LiTMUS trials). The Bipolar CHOICE study randomized participants into lithium and quetiapine treatment groups, and LiTMUS randomized participants into lithium with other personalized therapy (OPT), and non-lithium OPT only. Treatment lasted 24 weeks for all groups, and glucose levels were measured before and after the treatment period. Both studies reported no significant changes to FPG in the lithium only treatment arms, while the quetiapine-treated group had a marked increase in FPG ($p < 0.05$). A third study, conducted by McIntyre et al. (2011), conducted a 52-week randomized, placebo-controlled, double-blind study of lithium-treated and aripiprazole-treated patients and also found no overall differences in FPG levels between each treatment group, and the placebo group. Participants were randomized to the lithium intervention group, the aripiprazole intervention group, and the placebo group. All groups received treatment for three weeks, but lithium and aripiprazole groups received treatment for 12 weeks and some participants continued treatment for 40 weeks as per investigator judgement. Measurements were obtained prior to treatment, at three weeks, 12 weeks at completion, and 40 weeks post-completion. At week 52, none of the reported changes from baseline glucose levels were significant.

Vestergaard & Schou (1987) investigated the impact of long-term lithium treatment on glucose metabolism and Type 2 diabetes mellitus (T2DM) risk in their longitudinal study, and found no significant change to FPG after six years of lithium treatment. About half (49.1%) of study participants were started on lithium and their FPG levels were measured after 6 months of treatment, one year of treatment, and every year after for six years. The average FPG level of the intervention group at the start of the study was not significantly different from the average FPG level of the intervention group at year six. Moreover, the authors concluded that lithium treatment did not present an elevated risk for developing T2DM for manic-depressive patients.

One study, conducted by Agbayewa (1982), reported lower FPG levels in the first year of lithium treatment, with this effect diminishing by the third year of treatment. Agbayewa (1982) retroactively reviewed the charts of 43 bipolar patients followed at a psychiatric center in Saskatchewan, Canada. 22 patients were treated with lithium carbonate at the time of study while 21 age and sex-matched participants had never received lithium carbonate therapy. At baseline, the lithium group's mean FPG level and the control group's mean FPG level were not significantly different ($0.1 > p > 0.05$). At one year, the lithium-treated group demonstrated a reduction in mean FPG whereas the control group demonstrated an increase, and this difference was found to be significant ($p < 0.05$). This effect was reduced by the three year chart review, and authors reported there was no longer a significant difference in FPG between groups ($0.5 > p > 0.1$).

Studies of Other Metrics of Serum Glucose

Studies which investigated other glucose metrics, such as mean serum glucose, found lithium to be associated with elevated glucose, or found no significant effect. Tondo et al. (2017) and Mellerup et al. (1983) found that lithium treatment was associated with elevated mean serum glucose levels in patients with BD. Tondo et al. obtained participant data retrospectively using the databases of 12 international mood-disorder clinics. Participants underwent blood testing at regular intervals throughout the duration of lithium treatment. Glucose measurements at the beginning of lithium treatment were compared to measurements taken at various durations of lithium treatment using rANOVA testing. Mean serum glucose levels rose significantly throughout the duration of lithium treatment ($p < 0.001$). This increase was particularly pronounced after 6-10 years of lithium treatment, and serum glucose levels continued to rise for more than 31 years after the start of treatment. No healthy comparators were included.

Mellerup et al.'s (1983) study included a lithium-treated group of manic-depressive patients, manic-depressive psychiatric controls who were not treated with lithium, and healthy controls without psychiatric diagnoses. Groups were age and gender controlled and spent 24 hours in a Danish research department, where they were fed a standardized diet and had blood drawn serially 17 times. The lithium-treatment duration ranged between six months to nine years (median 4.5 years). The authors found that mean glucose levels for participants who were depressed and non-depressed did not different significantly; however, lithium-treated patients with depression had significantly lower mean glucose levels than non-depressed lithium-treated patients ($p < 0.01$). The non-depressed subgroup of lithium-treated patients had somewhat higher max glucose ($p < 0.05$) and mean glucose ($p < 0.05$) than both the psychiatric controls and the healthy controls. Additionally, the lithium-treated group's minimum glucose was higher than healthy controls ($p < 0.05$). The authors concluded that lithium-treatment was associated with "slight persistent hyperglycemia."

In Pinna et al.'s (2020) retrospective cohort study of patients with BD or major depressive disorder (MDD) who had received lithium treatment for at least 12 months, mean glucose levels were 97 ± 25 mg/dL, and 26.7 % of participants were hyperglycemic (>100 mg/dL). However, Pinna et al. did not identify a correlation between serum lithium levels and glucose levels in participants (p=0.79, r=0.03, F=0.13), or between lithium treatment duration and glucose levels (p=0.41, e=0.06, F=0.71). The authors could not determine a causal relationship between lithium treatment and hyperglycemia due to confounding factors such as AAP use by participants.

Klumpers et al.'s (2004) cross-sectional study from the Netherlands found that plasma glucose levels of lithium-treated patients with BD or schizoaffective disorder were within a normal range. The mean duration of lithium treatment was 115 ± 89 months, and 77.5% of participants used lithium daily. Participants underwent testing for lithium serum levels and glucose among other metabolic measurements, and results were controlled for age, gender, and BMI. Although precise values were not provided for serum glucose, the authors report that all serum glucose values were within normal range, and they did not identify a significant relationship between glucose levels and lithium treatment duration.

One study, Domino et al. (1985), which analyzed the red blood cell (RBC) constituents of lithium-treated bipolar-affective patients, reported improbable results which authors determined to be an artifact of experimental difficulties. This study was not further considered.

Studies of Glucose Tolerance Tests

Three studies utilized glucose tolerance tests (GTTs) to investigate lithium's impact on metabolic markers, and all three reported lithium-treated patients to exhibit an increased glucose tolerance in the short-term. Both Vendsborg (1979) and Van der Velde and Gordon (1969) reported lithium to be associated with an increase in glucose tolerance in the short-term, but their conclusions on the long-term impacts of lithium on glucose tolerance differ. In Vendsborg's prospective study, an increase in glucose tolerance between 2-12 hours after lithium treatment was observed, but after three weeks of treatment this was no longer the case. GTTs were administered to participants prior to and the day after beginning lithium treatment. Lithium was administered after the first GTT, and an additional dose was administered as needed in the evening. The observed increase in mean serum glucose values and glucose tolerance between the levels prior to lithium treatment, two hours after treatment, and 12 hours after treatment was deemed significant ($p < 0.05$), with the exception of the change in serum glucose at 12 hours after treatment. Authors concluded that shortly after lithium treatment there was a significant increase in glucose tolerance irrespective of how long the patient had been taking lithium, but the period of time within which this increase could be observed decreased the longer the patient had received lithium.

In contrast, Van der Velde and Gordon (1969) concluded that in the short-term, glucose tolerance increased during the treatment periods of lithium carbonate and decreased after lithium carbonate withdrawal, but that this effect is more prominent with prolonged lithium exposure. At the beginning of this study, lithium carbonate was withheld from participants for two weeks at a time and readministered for two weeks at a time, twice over for a total of eight weeks. GTTs were completed and glucose levels recorded at one-half, one, and two-hour periods post-lithium administration, and summed to obtain glucose tolerance sums (GTS). The GTS values at the beginning of the study, when participants had been on lithium for at least a month, were consistently lower than in the two-week period of lithium carbonate re-introduction. GTS values in the second period of withholding lithium (the final two weeks of the study) were significantly higher than in the first withholding period (the third and fourth weeks of the study).

Shah et al. (1986) also found chronic lithium treatment to be associated with hypoglycemia–indicating an increased glucose tolerance–directly after GTTs. Bipolar patients who had been treated with lithium alone for six or more months at the time of testing, and a control group of patients who had previously met DSM-III criteria for BD but were currently in remission and not being treated pharmacologically, underwent GTTs. The mean serum glucose concentration for the intervention group was significantly lower than the mean serum glucose concentration for the control group ($P < 0.01$). The mean nadir glucose value for the intervention group was also significantly lower than the mean nadir glucose value for the control group ($P < 0.001$). Additionally, 7/9 of the lithium-treated participants experienced hypoglycemic symptoms, compared to 0 of the control participants. The authors determined an association between chronic lithium treatment and biochemical and symptomatic hypoglycemia during GTT.

Studies of Serum Insulin and Insulin Tolerance

Three studies investigated lithium's impact on insulin, and study findings differed greatly. Heninger & Mueller's (1970) prospective cohort study of patients expressing mania found lithium to have no significant effect on insulin or glucose parameters. All study participants were voluntarily hospitalized on a research ward to partake in this study. To assess the importance of clinical improvement in comparison to change stemming from the lithium carbonate treatment, participants were divided into a high improvement intervention group, and uncertain or poor improvement, based on improvement of manic-excitement symptoms. This study involved GTT and ITT testing performed on participants during a period of mania, and then a second test period during lithium carbonate treatment. In the GTTs, lithium carbonate treatment was associated with reduced glucose utilization for both improvement groups. The high improvement group's insulin levels were significantly higher during the treatment period than in the manic period at 20, 50 and 60 minutes after glucose injection ($p < 0.05$). For the ITTs, there was no difference in fasting glucose levels before and during lithium carbonate treatment. In the manic period, the high improvement group had lower glucose levels following insulin injection at 15 ($p < 0.05$), 30 ($p < 0.02$) and 120 minutes ($p < 0.05$) compared to glucose levels observed during the lithium treatment period. Results from both the GTTs and the ITTs demonstrate an association between lithium carbonate treatment and relative insulin resistance, but authors conclude that this relative resistance is likely due to increased sensitivity to insulin in mania and not due to lithium carbonate treatment.

In their cross-sectional study, Mellerup et al. (1972) found increased serum insulin levels in lithium-treated patients compared to untreated patients. Patients with a history of manic-depressive psychosis, but who were currently euthymic and had been for several months, were grouped based on whether they were receiving lithium therapy or no medications. 16 healthy controls were also recruited. The lithium-treated patients had received lithium for between six months and eight years. The lithium-treated group was further divided into patients who had a “marked increase” in weight (>5kg) over the course of their lithium treatment and those who had not. Mean fasting serum insulin levels were similar in the both lithium-treated groups and were slightly elevated relative to the healthy controls. However, they were nearly double the untreated manic-depressive group ($p < 0.01$), which is the most comparable control. The authors suggest that lithium may be responsible for insulin and carbohydrate metabolism changes, and that the observed weight increases in some (but not all) lithium treated patients may be secondary to this metabolic change.

In their prospective study testing the impacts of lithium therapy on insulin and other biomarkers in psychotic and manic-depressive patients, Seven et al. (1993) found that lithium treatment may be responsible for lowering insulin levels. Patients with psychosis (manic-depressive or schizophrenic) received either lithium and neuroleptic therapy or only neuroleptic therapy. Fasting blood samples were drawn from all patients prior to receiving their specified treatment and on the 15th day of treatment. Authors found that plasma insulin levels in the lithium and neuroleptic treatment group had decreased significantly by the 15th day of therapy compared to the decrease in plasma insulin levels observed in the neuroleptic-only treatment group ($p = 0.001$). There was a weak, insignificant correlation between the amount of lithium found in plasma and the changes in metabolic parameters ($r = -0.31$ for insulin), which may be attributed to the small sample size.

Discussion

Fasting plasma glucose was one of the most extensively researched biomarkers in our review. FPG is a particularly useful metric for understanding the effects of lithium given that its measurement is highly standardized, and that it is commonly ordered to monitor metabolic changes in patients. Elevated FPG is also a diagnostic finding of diabetes mellitus (American Diabetes Association Professional Practice Committee, 2023). Several of the reviewed studies suggest that lithium treatment, in different forms, does not directly impact FPG levels in bipolar patients, most of which monitored FPG for up to one year of lithium treatment. Vestergaard & Schou (1987) followed lithium-treated patients up to 6 years and did not report a significant change in FPG longitudinally from year 0 to year 6, including at the intermediate timepoints (6 months, 1 year, and every subsequent year). There was also no significant difference in FPG between lithium-treated patients and the control group. However, despite a very large total participant size (n=460), only 15 participants remained in the study by year 6. Given the small long-term enrollment and the age of the study, additional studies would be helpful to confidently establish the long-term effects of lithium treatment on FPG.

An alternate finding of FPG comes from Agbayewa's (1982) retroactive chart-review, where the lithium-treated group had a significant decrease in FPG after 1 year of treatment, whereas the control group had slightly increased FPG (N.S.). By 3 years of treatment, fasting glucose levels had stabilized and were no longer significantly different from their baseline. There were also no significant differences between groups. However, the study did not control for key conflating variables such as mood stability, dietary changes, or BMI, and much of the original data is not publicly available. These factors could potentially explain the finding of decreased FPG: for example, patients with better mood stability due to lithium's clinical effects may have better eating habits (Martin et al., 2016; McDonald et al., 2019), which could lead to the reduction in FPG. Moreover, this study was not randomized and is substantially smaller (n=43) than the several large RCTs which did not find an association between lithium treatment and fasting glucose decreases (Köhler-Forsberg et al., 2023; Kuperberg et al., 2022; McIntyre et al., 2011;

Vestergaard and Schou, 1987). Interestingly, this study showed a stabilization of FPG levels in the lithium-treated group between years 1 and 3. This effect, as well as the possibility of a short-term decrease in FPG after initiation of lithium carbonate treatment, warrants further investigation. The balance of reviewed evidence supports a conclusion that lithium therapy does not cause lasting changes to FPG.

Several studies examined other metrics of serum glucose, such as random glucose or 24-hour mean glucose, and generally these studies have less concordance with one another. This inconsistency may be inherent to the metrics used—for example, random serum glucose varies significantly throughout the day and postprandially. Moreover, the reviewed studies did not standardize the method/time for measurement of random glucose. Additionally, 24-hour, maximum, and minimum glucose levels were assessed by Mellerup et al. (1983), but more research is needed to form sound conclusions on these longitudinal measures of glucose. Ultimately, the reviewed papers—Tondo et al. (2017), Mellerup et al. (1983), Pinna et al. (2020), Klumpers et al. (2004), and Domino et al. (1985)—do not paint a clear picture of how lithium affects other metrics of serum glucose.

Glucose Tolerance Tests (GTTs) were utilized by several studies to elucidate how lithium treatment affects the glucose tolerance of bipolar individuals. Glucose tolerance assesses the body's ability to dispense of a glucose load, largely via cellular uptake. It is intimately linked with insulin sensitivity and with insulin/glucagon regulation by pancreatic alpha and beta cells. GTTs are usually performed by taking a baseline glucose level in fasting state, then administering a large glucose dose (typically 75g), waiting a fixed period of time to allow for cellular uptake of glucose, and then serially measuring serum glucose at a fixed interval to assess the speed that glucose is dispensed from blood (World Health Organization, 1999). Vendsborg (1979) performed GTTs in patients taking lithium and concluded that lithium treatment increases glucose tolerance in the short-term. Van der Velde and Gordon (1969) also performed GTTs on lithium-carbonate patients (minimum treatment duration was 1 month) and also concluded that lithium carbonate increases glucose tolerance in the short-term. Further, these findings are consistent with Shah et al.'s (1986) finding that chronic lithium treatment is associated with symptomatic and biochemical hypoglycemia following GTTs. Compared to non-lithium-treated bipolar controls, lithium-treated patients had significantly lower mean and nadir glucose 5-hours after GTT. Overall, there were limited studies of glucose tolerance in lithium treated patients. These studies had fairly small sample size ($n \leq 19$) and were all conducted more than 40 years ago. Newer and larger scale research is needed to reliably assess glucose tolerance.

3 studies in this review focus specifically on evaluating insulin parameters—Heninger and Mueller (1970), Mellerup et al. (1972), and Seven et al. (1993). These studies generally do not suggest a causal relationship between lithium and changes to insulin homeostasis. Given the small study sizes and heterogeneity of assessed parameters, additional studies are needed to make these findings generalizable to clinical practice.

Strengths and limitations

To our knowledge, this is the first review to systematically appraise studies exploring the impact of lithium treatment on glucose and insulin parameters in bipolar persons. Our findings are limited by the lack of consistent criteria for BD diagnosis across studies, especially since many studies were conducted prior to the addition of BD to the DSM. As such, the 16 included studies were closely examined and reviewed by multiple authors to determine if the studies' inclusion criteria for participants were sufficiently aligned with our population of interest. It was infeasible to conduct a meta-analysis due to the heterogeneity of collected data across studies. Finally, many of the study populations are female dominated, although males are sufficiently represented across the study groups.

Importance and Future Directions for Research

There is ample evidence, across a wide age range and varying treatment durations, suggesting that lithium does not significantly impact FPG. The findings of this review suggest that the common practice of prescribing lithium for short- and long-term treatment of BD need not be discontinued out of concern for metabolic parameters, particularly FPG. However, there are significantly fewer papers, generally of smaller and more homogenous sample sizes, for the other metabolic markers of interest. This limits our ability to make generalizable conclusions with respect to serum insulin, glucose tolerance, and other serum glucose markers. Further research is needed on lithium's impact on other metrics of glucose. Additionally, the existing literature which utilizes GTTs and ITTs to investigate lithium's impact on glucose and insulin tolerance have small sample sizes, high heterogeneity, and were all conducted more than 30 years ago. Therefore, newer, larger studies which have consistent parameters and utilize ITTs and GTTs to assess lithium's metabolic effects are also needed to support the creation of robust clinical guidelines for lithium prescription.

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CRediT Author Statement

David Kanter-Eivin: Writing - Original Draft, Conceptualization, Project Administration, Investigation; Konstantin Blinkov: Conceptualization, Visualization, Investigation, Writing - Review & Editing; Sadie MacDonald: Writing - Original Draft, Investigation; Jada Gibson: Investigation, Writing - Review & Editing; Sarah Pimenta: Investigation, Writing - Review & Editing; Kayla Teopiz: Writing - Review & Editing, Supervision, Conceptualization; Angela Kwan: Writing - Review & Editing; Gia Han Le: Writing - Review & Editing; Sabrina Wong: Writing - Review & Editing; Roger McIntyre: Supervision, Conceptualization, Writing - Review & Editing.

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